Knowledge Representation

Lecture 14: Rough Sets

dr Anna Maria Radzikowska

Wydział Matematyki i Nauk Informacyjnych

Pokój 504, gmach MiNI

E–mail: A.Radzikowska@mini.pw.edu.pl

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- Rough set–style data analysis provide efficient tools for dealing with these problems.

Preliminaries

Let X be a non–empty set and let $R \subseteq X \times X$ be a binary relation on X.

- ***Equivalence relation*** — satisfies:
  - *reflexivity*: xRx for every $x \in X$
  - *symmetry*: $xRy \Longrightarrow yRx$ for all $x, y \in X$
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- ***Partition*** — the set $\Pi = \{X_1, \ldots, X_k\}$ of non–empty subsets of X satisfying:
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- ***Equivalence class***: — the set $[x]_R = \{y \in X : xRy\}$, $x \in X$.
- Each equivalence relation $R \subseteq X \times X$ generates the partition $X/_R = \{X_1, \ldots, X_k\}$, where each X_i is its equivalence class.

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- for any $a \in \mathcal{A}t$, V_a is a set of values (domain) of the attribute a
- each attribute $a \in \mathcal{A}t$ is a mapping $a : Ob \to V_a$; for $x \in Ob$, $a(x)$ is the value of the object x wrt the attribute a

Information system (cont.)

An information system $\Sigma = (Ob, \mathcal{A}t, \{V_a : a \in \mathcal{A}t\})$ can be represented by a table as below.

$Ob/\mathcal{A}t$	a_1	a_2	$\ldots$	a_k
x_1	$a_1(x_1)$	$a_2(x_1)$	$\ldots$	$a_k(x_1)$
x_2	$a_1(x_2)$	$a_2(x_2)$	$\ldots$	$a_k(x_2)$
$\vdots$	$\vdots$	$\vdots$	$\vdots$	$\vdots$
x_n	$a_1(x_n)$	$a_2(x_n)$	$\ldots$	$a_k(x_n)$

Information system — Example 14.1

Let $Ob = \{x_1, \ldots, x_7\}$, $\mathcal{A}t = \{\textbf{\textsf{A}}nimality, \textbf{\textsf{S}}ize, \textbf{\textsf{C}}olor\}$,
$V_A = \{$bear, dog, cat, horse$\}$, $V_S = \{$small, medium, large$\}$,
$V_C = \{$brown, black$\}$.

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object	*Animality*	*Size*	*Color*
x_1	bear	small	black
x_2	bear	medium	black
x_3	dog	large	brown
x_4	cat	small	black
x_5	horse	medium	black
x_6	horse	large	black
x_7	horse	large	brown

Indiscernibility

Let $\Sigma = (Ob, \mathcal{A}t, \{V_a : a \in \mathcal{A}t\})$ be an information system, let $A \subseteq \mathcal{A}t$ be a set of attributes, and let $x, y \in Ob$ be two objects. We say that x and y are ***A–indiscernible***, in symbols $x \underset{A}{\sim} y$, iff for every $a \in A$, $a(x) = a(y)$.

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Each equivalence class $[x]_{\underset{A}{\sim}}$ consists of objects indiscernible wrt all properties $a \in A$ and is called a ***granule***. Every granule has a unique characterization. Any subset $X \subseteq Ob$ can be characterized by means of these granules.

Characterization of granules

Let $Ob/_{\underset{A}{\sim}} = \{\Gamma_1, \ldots, \Gamma_m\}$ be the sey of all granules (equivalence classes of $\underset{A}{\sim}$). Each granule Γ can be characterized by the following expression:

$$\boxed{ch(\Gamma) = a_{i_1} \text{ IS } v_{i_1} \ \wedge \ a_{i_2} \text{ IS } v_{i_2} \ \wedge \ \ldots \ \wedge \ a_{i_p} \text{ IS } v_{i_p}}$$

where for every $j = i_1, \ldots, i_p$,

- $a_j \in \mathcal{A}t$, and
- $v_j \in V_{a_j}$ — a value of the attribute a_j.

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- $a_j \in \mathcal{A}t$, and
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The set $\Gamma_1 \cup \Gamma_2 \cup \ldots \cup \Gamma_i$ of granules can be then characterized by:

$$\boxed{ch(\Gamma_1) \text{ OR } ch(\Gamma_2) \text{ OR } \ldots \text{ OR } ch(\Gamma_i)}$$

Example 14.1 (cont.)

object	*Animality*	*Size*	*Color*
x_1	bear	small	black
x_2	bear	medium	black
x_3	dog	large	brown
x_4	cat	small	black
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x_6	horse	large	black
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Let $A = \{Animality, Color\}$.

$\Gamma_1 = \{x_1, x_2\}$ (black bear)

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$A = \{Animality, Color\}$.

- $\Gamma_1 = \{x_1, x_2\}$: *black bear*

  $Animality$ IS $bear \ \wedge \ Color$ IS $black$.

- $\Gamma_2 = \{x_3\}$: *brown dog*

  $Animality$ IS $dog \ \wedge \ Color$ IS $brown$.

- $\Gamma_3 = \{x_4\}$: *black cat*

  $Animality$ IS $cat \ \wedge \ Color$ IS $black$.

- $\Gamma_4 = \{x_5, x_6\}$: *black horse*

  $Animality$ IS $horse \ \wedge \ Color$ IS $black$.

- $\Gamma_5 = \{x_7\}$: *brown horse*

  $Animality$ IS $horse \ \wedge \ Color$ IS $brown$.

Rough Approximations

For any $A \subseteq \mathcal{A}t$ define two operations $\underline{A}, \overline{A} : 2^{Ob} \longrightarrow 2^{Ob}$ as follows: for every $X \subseteq Ob$,

$$\boxed{\begin{aligned} \underline{A}(X) &= \{x \in Ob : [x]_{\underset{A}{\sim}} \subseteq X\} \\ \overline{A}(X) &= \{x \in Ob : [x]_{\underset{A}{\sim}} \cap X \neq \emptyset\} \end{aligned}}$$

$\underline{A}(X)$: *A–lower rough approximation of* X

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Intuitively:

$x \in \underline{A}(X)$: x *certainly belongs to* X

$x \in \overline{A}(X)$: x *possibly belongs to* X.

Rough Set

Let $\Sigma = (Ob, \mathcal{A}t, \{V_a : a \in \mathcal{A}t\})$ be an information system, let $A \subseteq \mathcal{A}t$, and let $X \subseteq Ob$. We say that X is

- A–***definable*** iff $\underline{A}(X) = \overline{A}(X)$
- A–***rough*** otherwise.

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Note that both $\underline{A}(X)$ and $\overline{A}(X)$ are unions of granules.

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The set $Bd_A(X) = \overline{A}(X) \setminus \underline{A}(X)$ is called a ***A–boundary region***.

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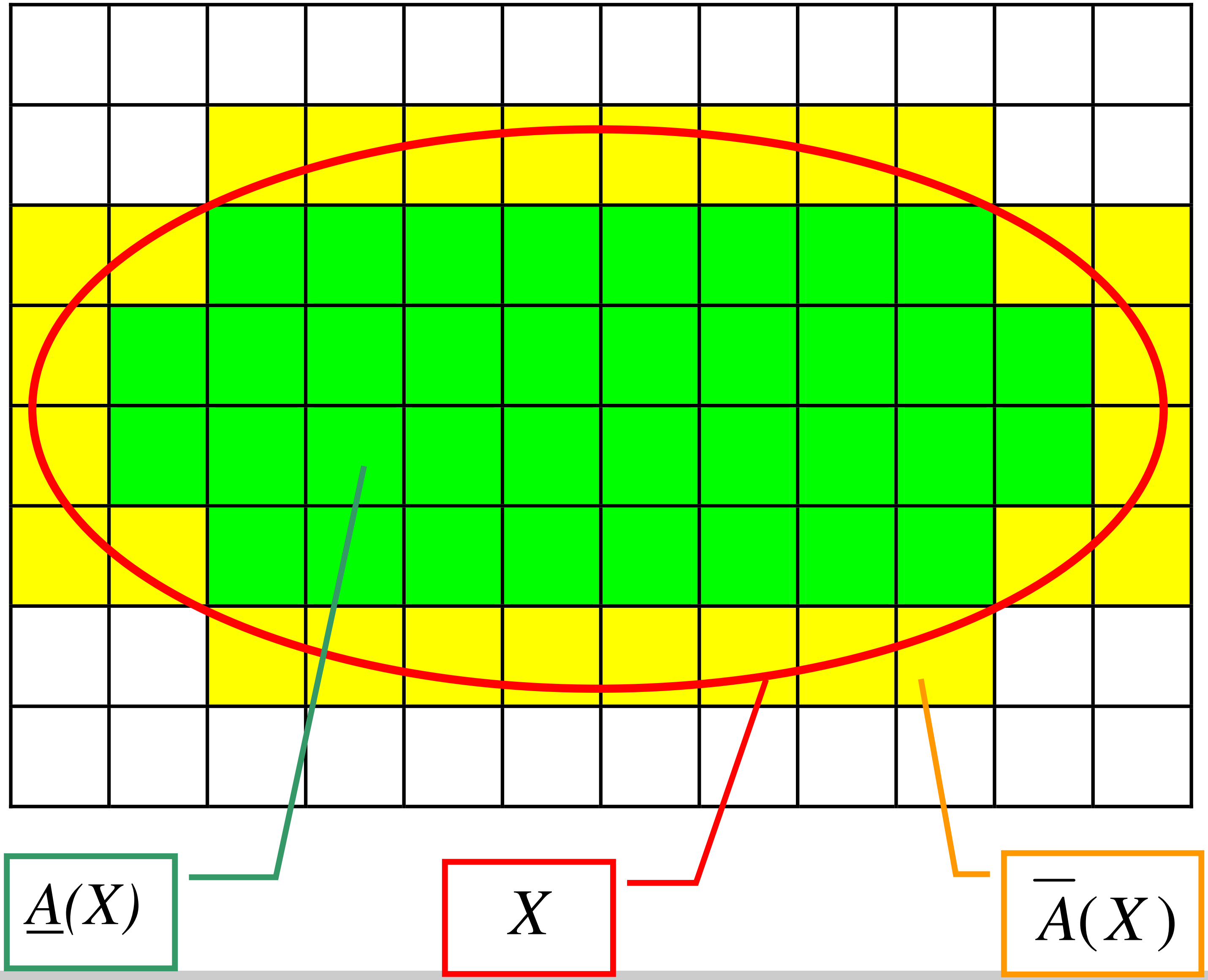
By an ***A–rough set*** we mean a pair (X_1, X_2) of subsets of Ob such that $X_1 = \underline{A}(X)$ and $X_2 = \overline{A}(X)$ for some $X \subseteq Ob$.
(X_1, X_2) is called a ***rough set*** iff it is an A–rough set for some $A \subseteq \mathcal{A}t$.

Rough set (cont.)



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Let $X_1 = \{x_1, x_2, x_3\}$

X_1 is A–definable and $X_1 = \Gamma_1 \cup \Gamma_2$

Characterization:

$Animality$ IS $bear$ $\wedge$ $Color$ IS $black$

OR $Animality$ IS dog $\wedge$ $Color$ IS $brown$.

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Characterization:

$Animality$ IS *bear* $\wedge$ $Color$ IS *black*

OR $Animality$ IS *dog* $\wedge$ $Color$ IS *brown*.

Let $X_2 = \{x_1, x_2, x_5\}$.

X_2 is not A–definable:

$\underline{A}(X_2) = \Gamma_1, \quad \overline{A}(X_2) = \Gamma_1 \cup \Gamma_4$.

Characterization of X_2:

black bear (certainly)

black bear OR black horse (possibly)

Note: $x_5 \in X_2$, $x_6 \notin X_2$, $x_5, x_6 \in \Gamma_4$.

Basic properties

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- $X \subseteq X' \Longrightarrow \underline{A}(X) \subseteq \underline{A}(X')$ & $\overline{A}(X) \subseteq \overline{A}(X')$.
- $\underline{A}(X) \subseteq X \subseteq \overline{A}(X)$
- $\underline{A}(X \cap X') = \underline{A}(X) \cap \underline{A}(X')$
  $\underline{A}(X \cup X') \supseteq \underline{A}(X) \cup \underline{A}(X')$

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Example 14.2

Let $\Gamma_1 = \{x_1, x_2\}$, $\Gamma_2 = \{x_3, x_4, x_5\}$, $\Gamma_3 = \{x_6, x_7\}$.

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$$\begin{array}{ll} \overline{A}(X \cup X') = \Gamma_1 \cup \Gamma_2 \cup \Gamma_3 & \overline{A}(X \cap X') = \emptyset \\ \overline{A}(X) \cup \overline{A}(X') = \Gamma_1 \cup \Gamma_2 \cup \Gamma_3 & \overline{A}(X) \cap \overline{A}(X') = \Gamma_2 \\ \overline{A}(X \cup X') = \overline{A} \cup \overline{A}(X') & \overline{A}(X \cap X') \subseteq \overline{A} \cap \overline{A}(X'). \end{array}$$

Basic properties (cont.)

- ***Duality***:

$$\underline{A}(X) = -\overline{A}(-X)$$
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($-X$ stands for the complementation of X).

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  ($-X$ stands for the complementation of X).

- $A \subseteq A' \Longrightarrow \underset{A'}{\sim} \subseteq \underset{A}{\sim}$.
  The more extensive the set of attributes is, the finer the granularization is.
  In consequence, taking $\mathcal{A}t$ we obtain the finest granularization.

Rough sets (cont.)

Let $\Sigma = (Ob, \mathcal{A}t, \{V_a : a \in \mathcal{A}t\})$ be given, $A \subseteq \mathcal{A}t$, and let $X \subseteq Ob$. Assume that X is not A–definable. Then we have the following four cases:

- $\underline{A}(X) \neq \emptyset$ and $\overline{A}(X) \neq Ob$.
  We can determine features which objects of X possess and ones which they do not have (i.e. some examples and some counterexamples).

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- $\underline{A}(X) \neq \emptyset$ and $\overline{A}(X) \neq Ob$.
  We can determine features which objects of X possess and ones which they do not have (i.e. some examples and some counterexamples).
- $\underline{A}(X) \neq \emptyset$ and $\overline{A}(X) = Ob$.
  We can determine features that objects of X have, but no features which they do no possess (only examples, no counterexamples).

Rough sets (cont.)

- $\underline{A}(X) = \emptyset$ and $\overline{A}(X) \neq \emptyset$.
  We can only determine features, which objects of X do not have (only counterexamples).

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- $\underline{A}(X) = \emptyset$ and $\overline{A}(X) \neq \emptyset$.
  We can only determine features, which objects of X do not have (only counterexamples).
- $\underline{A}(X) = \emptyset$ and $\overline{A}(X) = Ob$.
  We cannot give any features, which objects of X possess, and no features, which they do not possess.

Learning from examples

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- Let $\Sigma = (Ob, \mathcal{A}t, \{V_a : a \in \mathcal{A}t\})$ be given. The set Ob is now viewed as the set of *examples*.
- By a ***concept*** we mean an arbitrary non–empty subset $\mathcal{C}$ of examples.

Learning from examples

- Let $\Sigma = (Ob, \mathcal{A}t, \{V_a : a \in \mathcal{A}t\})$ be given. The set Ob is now viewed as the set of *examples*.
- By a ***concept*** we mean an arbitrary non–empty subset $\mathcal{C}$ of examples.
- Our task is to give the most precise characterization of $\mathcal{C}$ basing on the information in Σ.

Learning from examples (cont.)

Given $\Sigma = (Ob, \mathcal{A}t, \{V_a : a \in \mathcal{A}t\})$ and $\mathcal{C} \subseteq Ob$, our task is:

- Find the smallest set $A \subseteq \mathcal{A}t$ of attributes such that $\mathcal{C}$ is A–definable.

Learning from examples (cont.)

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- If $\mathcal{C}$ is not $\mathcal{A}t$–definable, then determine the smallest set $A \subseteq \mathcal{A}t$ such that for every $B \subseteq \mathcal{A}t$ it holds:
  - $\underline{B}(\mathcal{C}) \subseteq \underline{A}(\mathcal{C})$
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In other words, we aim to determine:

- the exact characteristics of the concept $\mathcal{C}$, provided that it is possible; otherwise
- a set of attributes determining, as precisely as possible, the approximation of $\mathcal{C}$.

Example 14.1 (cont.)

object	*Animality*	*Size*	*Color*
x_1	bear	small	black
x_2	bear	medium	black
x_3	dog	large	brown
x_4	cat	small	black
x_5	horse	medium	black
x_6	horse	large	black
x_7	horse	large	brown

Example 14.1 (cont.)

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x_1	bear	small	black
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Let $A = \{Animality, Size\}$.

$\Gamma_1 = \{x_1\}$ (small bear)

$\Gamma_2 = \{x_2\}$ (medium bear)

$\Gamma_3 = \{x_3\}$ (large dog)

$\Gamma_4 = \{x_4\}$ (small cat)

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$\Gamma_6 = \{x_6, x_7\}$ (large horse).

Example 14.1 (cont.)

Let $A = \{Animality, Size\}$.

$$
\begin{array}{ll}
\Gamma_1 = \{x_1\} & \text{(small bear)} \\
\Gamma_2 = \{x_2\} & \text{(medium bear)} \\
\Gamma_3 = \{x_3\} & \text{(large dog)} \\
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\Gamma_5 = \{x_5\} & \text{(medium horse)} \\
\Gamma_6 = \{x_6, x_7\} & \text{(large horse).}
\end{array}
$$

Consider the concept *dangerous animal* corresponding to the set

$\mathcal{C} = \{x_1, x_2, x_3, x_6, x_7\}$.

$\underline{A}(\mathcal{C}) = \Gamma_1 \cup \Gamma_2 \cup \Gamma_3 \cup \Gamma_6$.

Example 14.1 (cont.)

Let $A = \{Animality, Size\}$.

$\Gamma_1 = \{x_1\}$ (small bear)
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$\underline{A}(\mathcal{C}) = \Gamma_1 \cup \Gamma_2 \cup \Gamma_3 \cup \Gamma_6$.

Characterization:

$Animality$ IS $bear$ $\wedge$ $Size$ IS $small$
OR $Animality$ IS $bear$ $\wedge$ $Size$ IS $medium$
OR $Animality$ IS dog $\wedge$ $Size$ IS $large$
OR $Animality$ IS $horse$ $\wedge$ $Size$ IS $large$

Example 14.1 (cont.)

- The expression:

$$\begin{array}{l} Animality \text{ IS } bear \ \wedge \ Size \text{ IS } small \\ \text{ OR } Animality \text{ IS } bear \ \wedge \ Size \text{ IS } medium \end{array}$$

is equivalent to $Animality$ IS $bear$, because all bears in Ob are either small or medium.

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- The expression:

$$\begin{array}{l} Animality \text{ IS } dog \;\wedge\; Size \text{ IS } large \\ \text{ OR } Animality \text{ IS } horse \;\wedge\; Size \text{ IS } large \end{array}$$

  is equivalent to $Size$ IS $large$, since all large animals in Ob are either dogs or horses.

Example 14.1 (cont.)

- The expression:

$$\begin{array}{l} \mathit{Animality}\ \text{IS}\ \mathit{bear}\ \wedge\ \mathit{Size}\ \text{IS}\ \mathit{small} \\ \ \text{OR}\ \mathit{Animality}\ \text{IS}\ \mathit{bear}\ \wedge\ \mathit{Size}\ \text{IS}\ \mathit{medium} \end{array}$$

  is equivalent to $\mathit{Animality}$ IS bear, because all bears in Ob are either small or medium.

- The expression:

$$\begin{array}{l} \mathit{Animality}\ \text{IS}\ \mathit{dog}\ \wedge\ \mathit{Size}\ \text{IS}\ \mathit{large} \\ \ \text{OR}\ \mathit{Animality}\ \text{IS}\ \mathit{horse}\ \wedge\ \mathit{Size}\ \text{IS}\ \mathit{large} \end{array}$$

  is equivalent to Size IS large, since all large animals in Ob are either dogs or horses.

- Finally, we obtain the following characterization of the concept *dangerous animal*:

$$\boxed{\mathit{Animality}\ \text{IS}\ \mathit{bear}\ \text{OR}\ \mathit{Size}\ \text{IS}\ \mathit{large}}$$

Partial decisions

- Let $\Sigma = (Ob, \mathcal{A}t, \{V_a : a \in \mathcal{A}t\})$ be given and assume that for some objects an expert cannot take the decision as to whether they coincide or do not with the new concept.

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  - $D^0 = \{x \in Ob : d(x) = 0\}$.

Learning from examples (cont.)

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Q1 *Does the expert's lack of knowledge have any influence on the learning process?*

Q2 *Does the expert's lack of knowledge can be detected?*

Learning from examples (cont.)

- The expert's incomplete decision d is inessential for the learning process iff we can learn the characterization of the new concept C (corresponding to this decision) regardless of how objects fom D^0 are classified, i.e. positively $(+)$ or negatively $(-)$.

Learning from examples (cont.)

- The expert's incomplete decision d is inessential for the learning process iff we can learn the characterization of the new concept C (corresponding to this decision) regardless of how objects fom D^0 are classified, i.e. positively $(+)$ or negatively $(-)$.
- This can only be the case iff for each granule $G \in Ob/_{\mathcal{A}t}$ such that $G \cap D^0 \neq \emptyset$, it holds $G \cap D^+ \neq \emptyset$ and $G \cap D^- \neq \emptyset$.

  In other words, if G contains elements classified positively and those clasified negatively, then regardless of the classification of the remaining elements (belonging to D^0) we are unable to state whether G characterizes the new concept (since it contains elements from D^-) or it does not characterize it (since it contains elements from D^+).

Learning from examples (cont.)

Assume that there is a granule $G = \{e_1, \ldots, e_n\}$ such that G contains only elements from D^0, say $e_1, \ldots, e_i$, and from D^+ (i.e. $e_{i+1}, \ldots, e_n$). In such a case,

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- ◇ if all $e_1, \ldots, e_i$ are positively classified, then G will characterize the new notion
- ◇ if some e_k, $k = 1, \ldots, i$, is such that $e_k \in D^-$, then $G \cap \underline{\mathcal{A}t}(D^+) = \emptyset$, but $G \subseteq \overline{\mathcal{A}t}(D^+)$. Hence G does not characterize the new concept.

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- ◇ if all $e_1, \ldots, e_i$ are positively classified, then G will characterize the new notion
- ◇ if some e_k, $k = 1, \ldots, i$, is such that $e_k \in D^-$, then $G \cap \underline{\mathcal{A}t}(D^+) = \emptyset$, but $G \subseteq \overline{\mathcal{A}t}(D^+)$. Hence G does not characterize the new concept.

The case when G contains only elements from D^0 and D^- is analogous.

Learning from examples (cont.)

Concluding, the expert's incomplete decision influences on the learning process whenever both the following conditions hold:

- $\underline{\mathcal{A}t}(D^0) = \emptyset$ — no granules consists only of elements from D^0
- $Bd_{\mathcal{A}t}(D^0) \subseteq Bd_{\mathcal{A}t}(D^+ \cup D^-)$ – each granule containing elements from D^0 contains both elements from D^+ and D^-.

In any other case the expert's decision has the influence on the learning process.

Discovering incomple decisions

Let $\Sigma = (Ob, \mathcal{A}t \cup \{d\}, \{V_a : a \in \mathcal{A}t)$ be given, let $A \subseteq \mathcal{A}$, and let $D^+, D^-, D^0 \subseteq Ob$ be determined as before.

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In the second case we can give characterizations of granules from $\underline{A}(D^0)$, i.e. point out characteristics, for which the expert cannot decide whether they coincide with the new concept or not.

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In the second case we can give characterizations of granules from $\underline{A}(D^0)$, i.e. point out characteristics, for which the expert cannot decide whether they coincide with the new concept or not.

In other cases the expert's incomplete decision cannot be discovered.

Thank you for your attention!

Any questions are welcome.